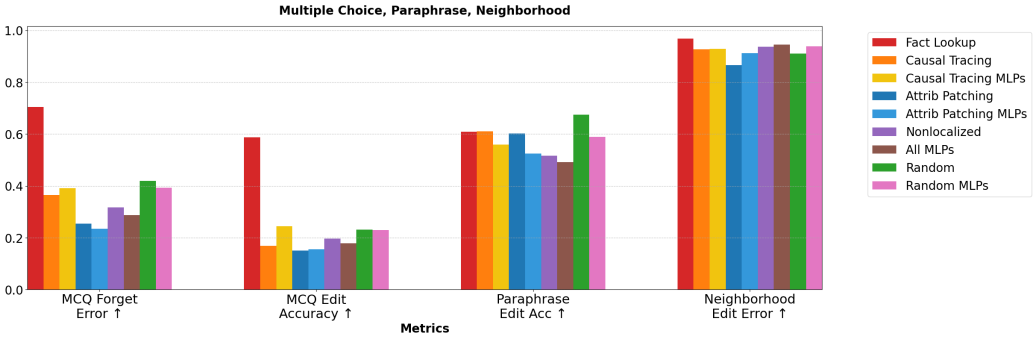

Figure 33: Adversarial Prompting results for Sequential-CounterFact-Editing, across all localizations.

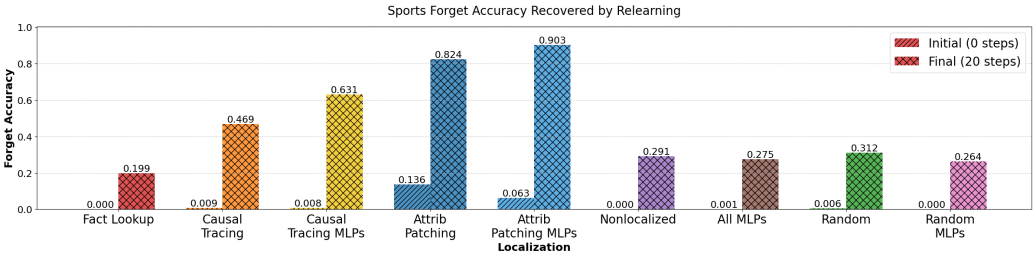

Figure 34: Relearning results for Sports-Athlete-Editing, across all localizations.

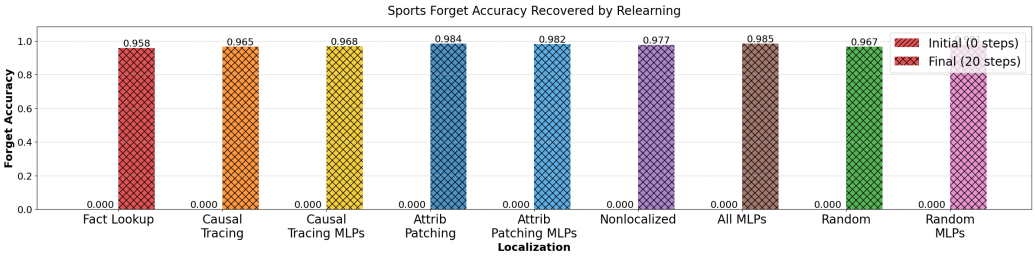

Figure 35: Relearning results for Full-Sports-Editing.

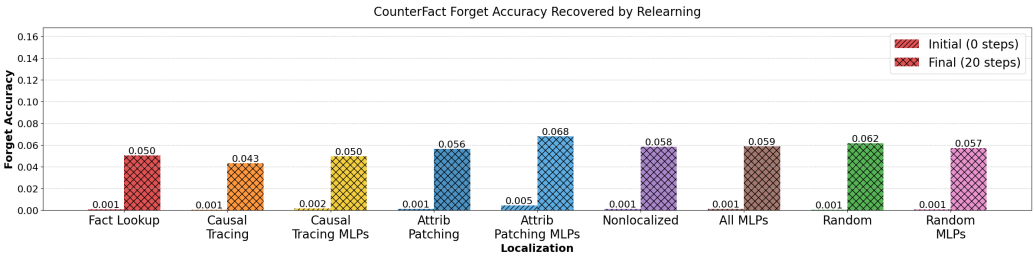

Figure 36: Relearning results for CounterFact-Editing.

four soft prompts. Soft prompt evaluations can be considered to be a more narrow form of few-shot finetuning, that is closer to searching for prompts that recover the model's knowledge.

We find limited soft prompt success: across most tasks and models, we don't recover much held-out forget set accuracy. On the Sports-Athlete-Editing, CounterFact-Editing, and Sequential-CounterFact-Editing tasks, Figures 42, 44 and 45 show that all localizations don't significantly improve in Forget Accuracy over random chance, or are about equal between localizations, after soft prompts are applied. In Figure 43, specifically for Gemma-2 on Sports-Athlete-Editing we see some

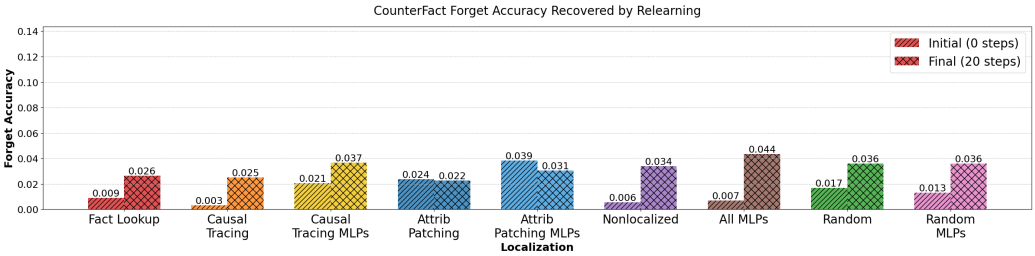

Figure 37: Relearning results for Sequential-CounterFact-Editing.

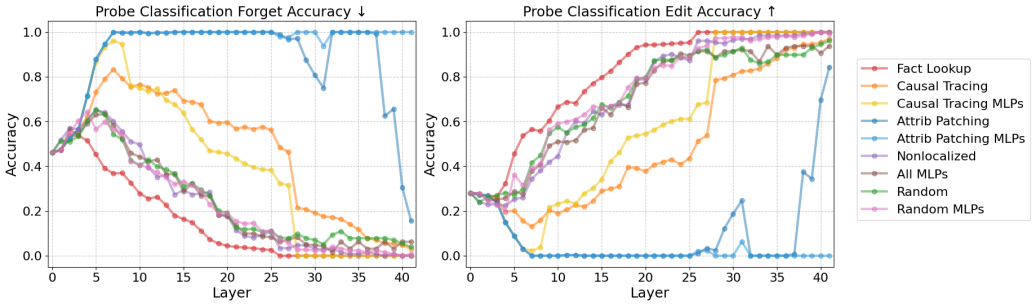

Figure 38: Linear probes applied to the forget set across all models, classifying model activations after various layers.

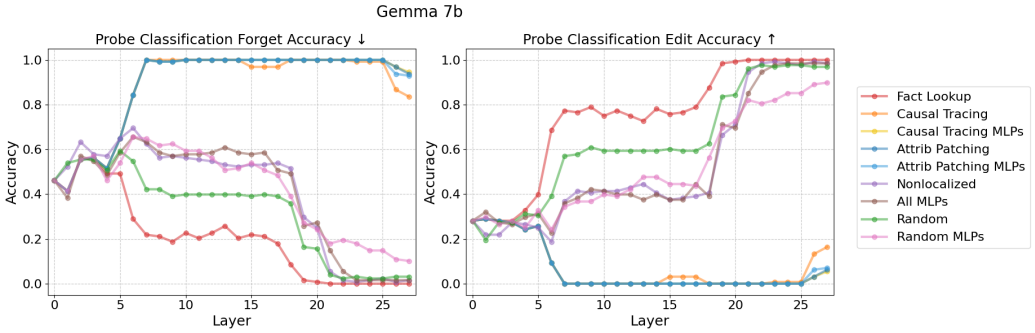

Figure 39: Linear probes applied to the forget set on Gemma-7B with 28 layers.

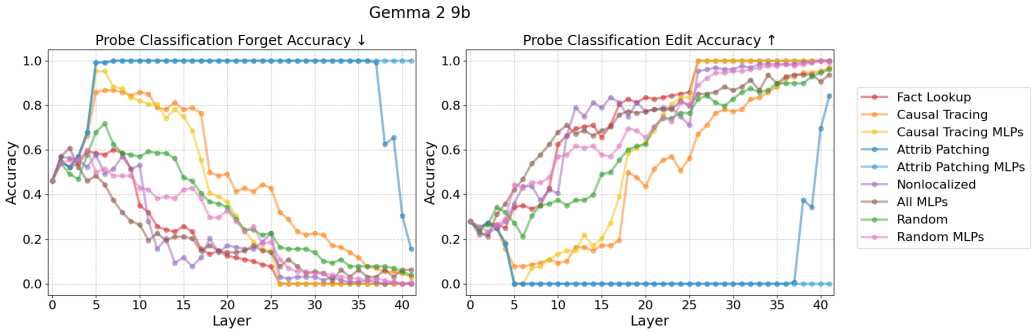

Figure 40: Linear probes applied to the forget set on Gemma-2-9b with 42 layers.

reasonable results with softprompts that are able to recover over 60% Forget Accuracy on OT localizations, while FLU, Nonlocalized, and All MLPs remain under 40% Forget Accuracy.

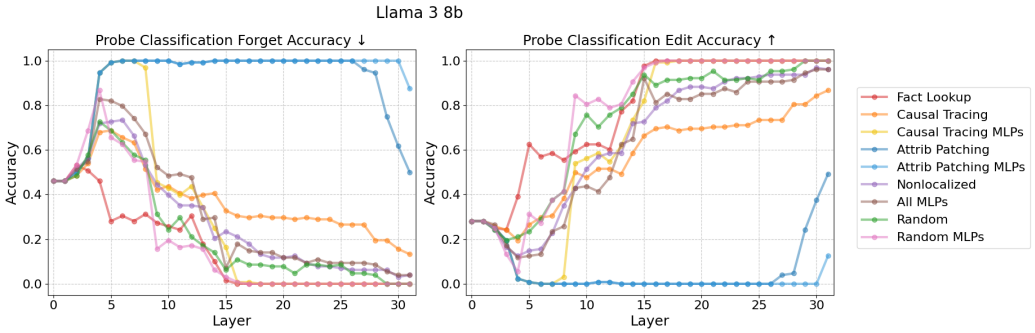

Figure 41: Linear probes applied to the forget set on Llama-3-8b with 32 layers.

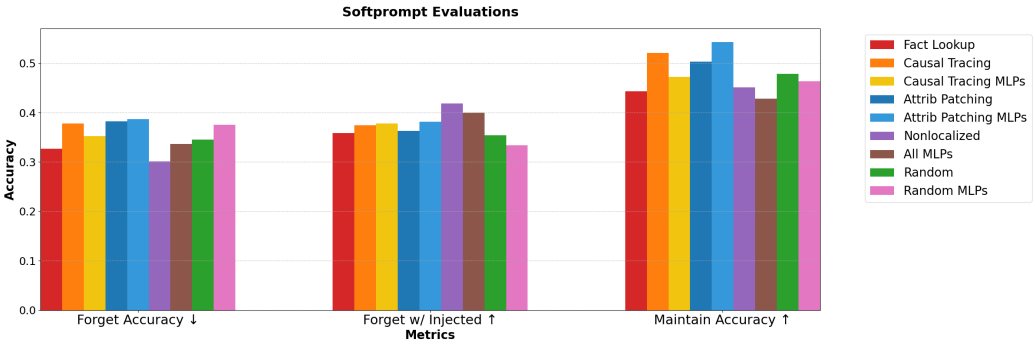

Figure 42: Metrics with soft prompts applied for Sports-Athlete-Editing, averaged over all models.

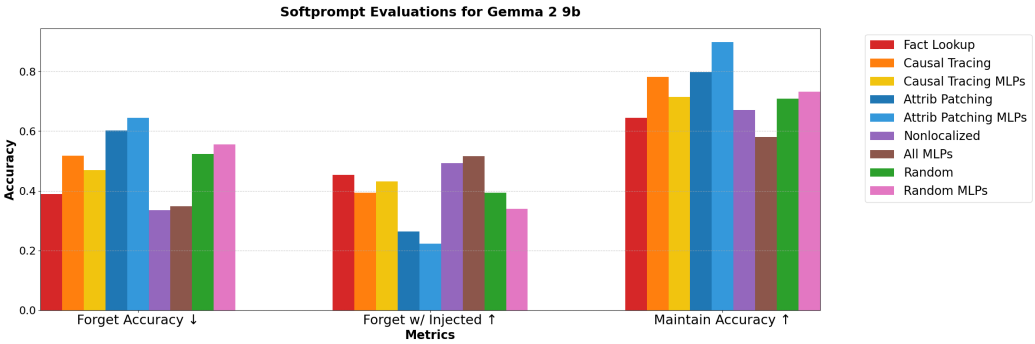

Figure 43: Metrics with soft prompts applied for Sports-Athlete-Editing for Gemma-2-9b.

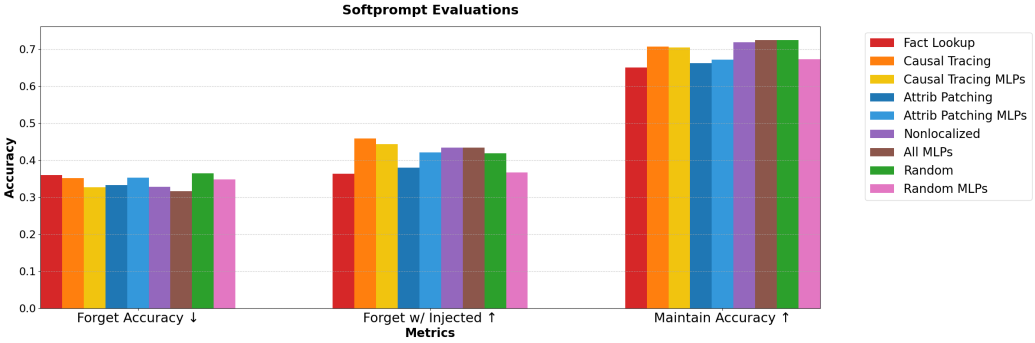

Figure 44: Metrics with soft prompts applied for CounterFact-Editing, averaged over all models.